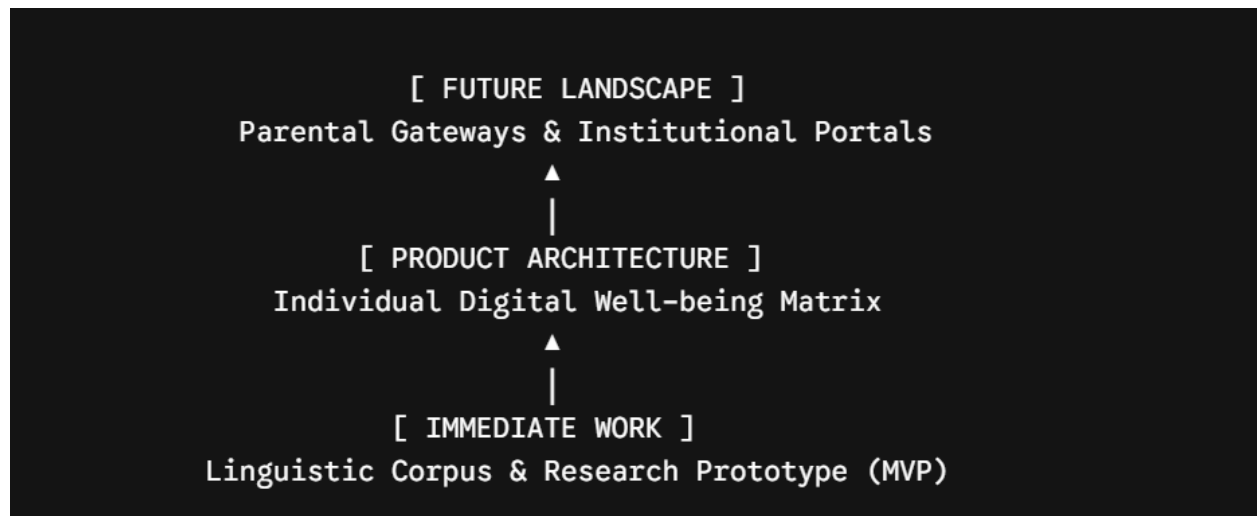


1. Product Vision & Strategy

Project Atlas is an explainable AI platform designed to analyze digital exposure and understand how young adults interact with masculine ideologies online. Rather than acting as a black-box detection tool, it maps social media feeds against established academic frameworks to provide transparent, multi-dimensional discourse metrics.

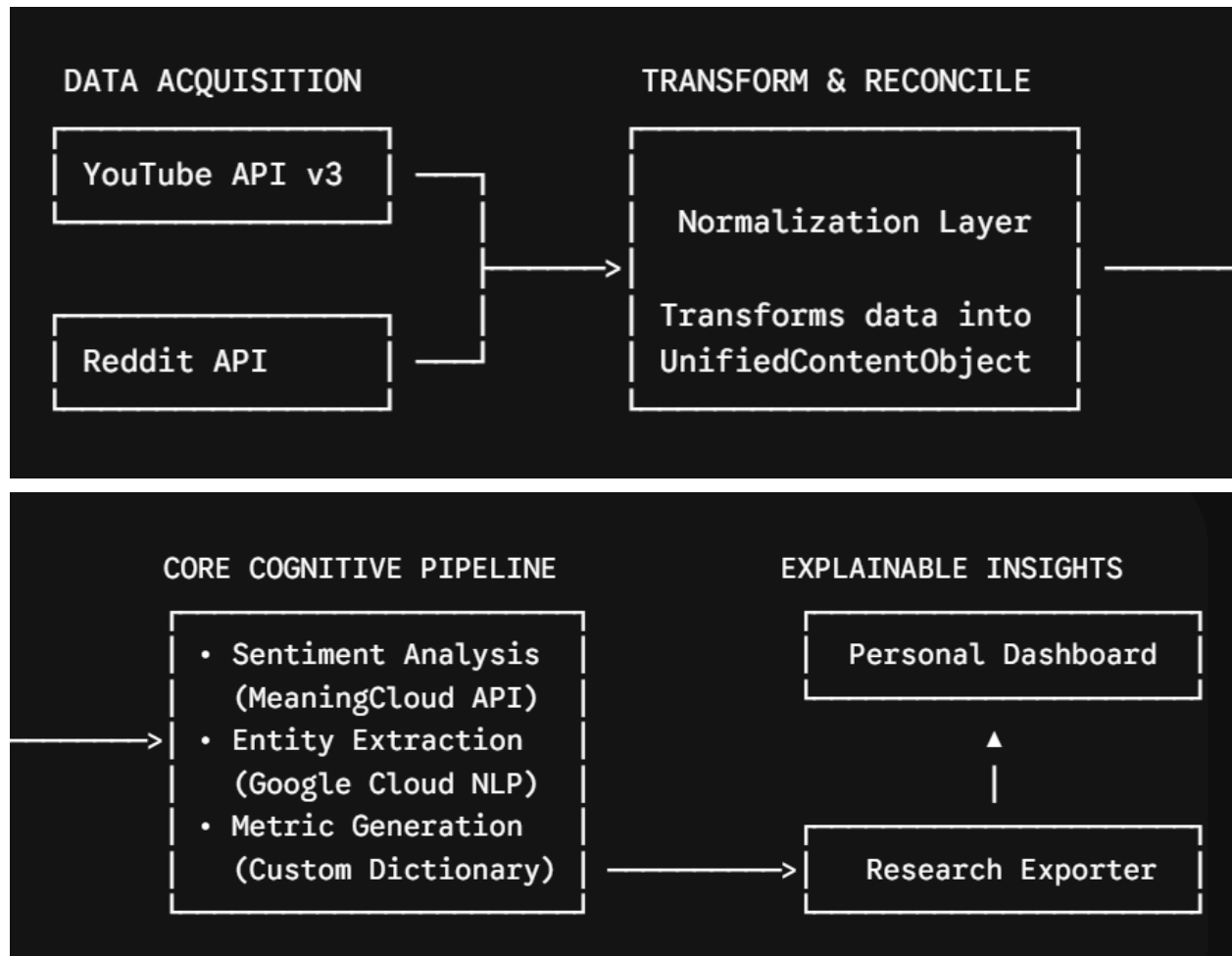
Core Strategic Focus



The Academic Alignment: The system directly implements **Connell's Hegemonic Masculinity Theory** and **Moral Foundations Theory** to measure multi-variable dimensions across physical, emotional, social, financial, and leadership vectors.

The MVP Approach: Your team will build the core platform architecture as if building the final software product, but **implement only the Research Prototype first**. This gives your advisor a working system immediately, leaves future expansion paths obvious, and keeps the potential patent viable.

2. High-Level System Architecture



3. The Central Architecture Paradigm: The Normalization Layer

To prevent the AI pipeline from being rewritten whenever a new platform connector is built, a strict **Normalization Layer** sits between data ingestion and analysis.

Plain-English Summary:

Instead of sending raw, messy data from YouTube or Reddit directly to the AI, this layer acts as an **universal translator**. It takes a YouTube watch log or a Reddit comment, strips out platform-specific clutter, and repacks it into an identical internal profile (containing a unique ID, the platform name, text type, raw body text, a timestamp, and anonymized user details).

Because the data is perfectly standardized, adding new networks down the road (like Instagram or X) will only require building a new plug-in rather than rebuilding the core AI engine.

4. Comprehensive 12-Module Engineering Breakdown

Module 1: Authentication & Identity Management

- **Plain-English Summary:** This module manages how a user logs into the system safely. It securely handles the "Sign In with Google" handshake and logs whether the user has read and checked the mandatory ethical privacy-consent agreements.
- **Stored Data:** It keeps a highly restricted database of a user's unique system ID, encrypted email, account name, and their active consent status.

Module 2: Platform Connections

- **Plain-English Summary:** This layer acts as the secure bridge between Project Atlas and a user's social accounts. It utilizes official, platform-approved verification standards (OAuth 2.0 tokens) so that the app can officially request read-only access to feeds without requiring or seeing user passwords.
- **Stored Data:** Keeps an encrypted list of security keys (tokens) linking the user's system ID to their connected YouTube and Reddit accounts, alongside automatic expiration timelines.

Module 3: Content Ingestion Node

- **Plain-English Summary:** This is the pipeline's extraction vehicle. It interfaces directly with the official YouTube Data API v3 and Reddit API to systematically pull down public comment threads, video descriptions, titles, metadata, and subreddit texts from the user's recent feed.
- **Stored Data:** Temporarily holds raw, unprocessed text feeds securely tied to an internal ID before passing them forward.

Module 4: The Ingestion & Cleaning Pipeline

- **Plain-English Summary:** Raw text from social media is filled with noise like typos, emojis, links, and formatting code. This module uses advanced linguistics toolkits (**NLTK and SpaCy**) to clean the text. It strips away emojis, breaks long paragraphs into individual sentences, and simplifies words to their root form (e.g., converting "grinding," "grinded," and "grinds" back to the core base word "grind") so the AI can analyze it accurately.

Module 5: Theoretical Knowledge Base Engine

- **Plain-English Summary:** This is the intellectual heart of the project. It houses your team's custom **Masculinity Dictionary**. It maps specific keywords, hashtags, and digital subculture slang to their respective academic categories under **Connell's Hegemonic Masculinity Framework** and **Moral Foundations Theory**.
- **Workflow:** Your team will research and tweak this entire dictionary inside an interactive scratchpad (**Jupyter Notebook**) before saving the completed rulebooks directly into the main system database.

Module 6: Core Cognitive Pipeline (The AI Engine)

- **Plain-English Summary:** Once text is cleaned and standardized, it runs through a series of specialized AI nodes. The system routes the text blocks through the **MeaningCloud API** to capture instantaneous emotional tone (Positive/Negative/Neutral) and leverages **Google Cloud Natural Language AI** to perform entity annotations—detecting names, specific online personalities, or major ideological terms driving the conversation.

Module 7: Core Metric & Measurement Matrix

- **Plain-English Summary:** This module calculates the numbers. It compiles the outputs from the AI engine and the masculinity dictionary, mapping the user's exposure onto 14 distinct dimensions (e.g., Physical Appearance, Emotional Stoicism, Social Dynamics, Financial Focus, Accountability, and Growth) using a sophisticated 0\text{-}100\$ scoring spectrum.

Module 8: Explainability & Evidence Engine

- **Plain-English Summary:** This is what makes your project stand out to academic judges. Rather than just handing out a random score, this module provides an **audit trail**. If a user scores an "85" in Aggression or Stoicism, this engine generates a card detailing exactly why, highlighting the specific keywords flagged, the exact comment clips they appeared in, and the system's mathematical confidence level.

Module 9: Narrative Recommendation Engine

- **Plain-English Summary:** Built to guide rather than shame, this module monitors daily text distributions. If it notices a heavy, repetitive exposure to hyper-isolated themes, it automatically crafts helpful, constructive, and educational wellness recommendations—suggesting the user expand their feed into topics like sports psychology, modern leadership, or mindfulness.

Module 10: Unified Storage Layer

- **Plain-English Summary:** The data vault. It uses a robust relational database engine (**PostgreSQL**) to manage all your structured application records, analytics runs, and user permissions securely. It is designed to scale effortlessly, allowing the integration of high-performance vector search extensions (**pgvector**) later on for deep contextual calculations.

Module 11: Application Program Interfaces (API Gateway)

- **Plain-English Summary:** The invisible central station. It acts as the traffic controller for the backend application, taking user commands from the web screen, routing them to the data cleaning or AI systems, and returning the polished analysis smoothly back to the user's interface.

Module 12: Visual Analytics Interface (The Front-End)

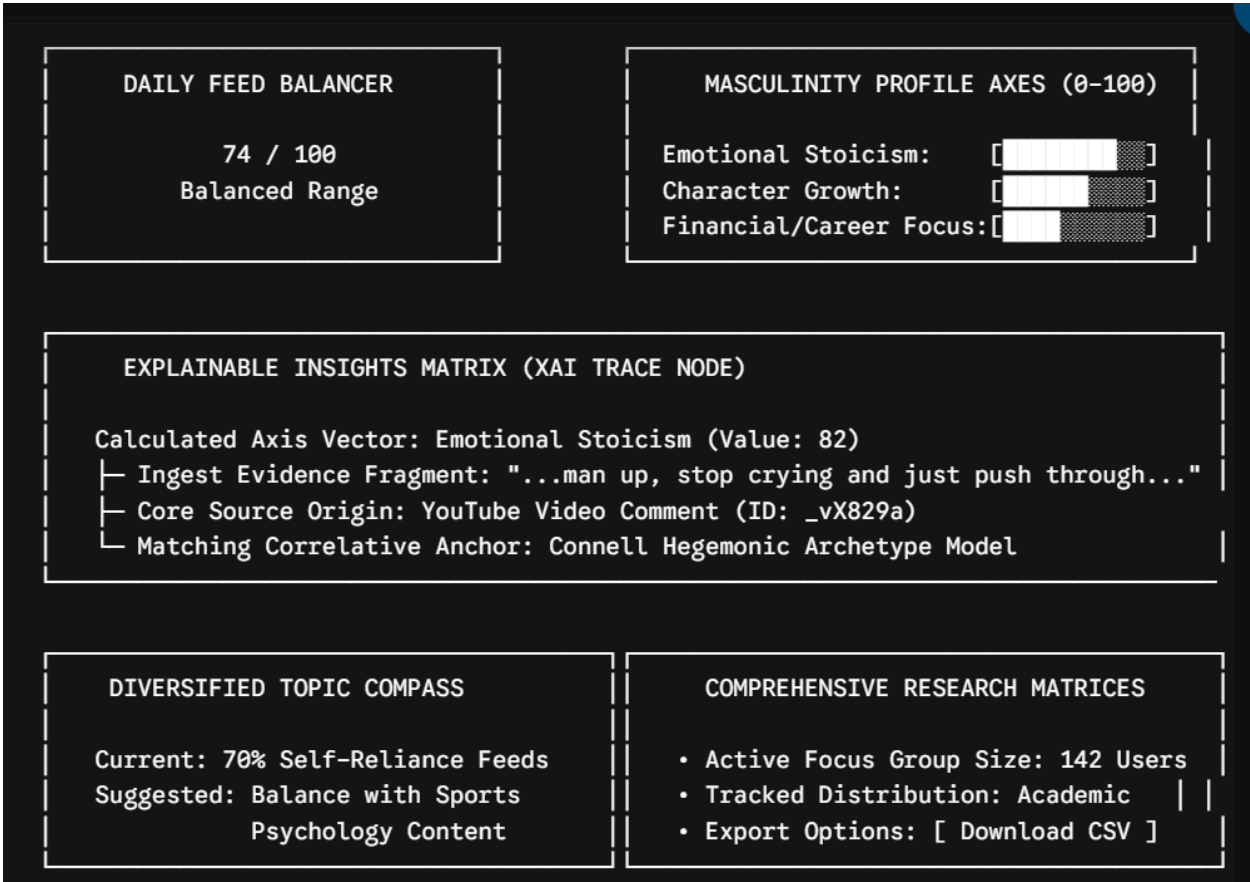
- **Plain-English Summary:** The outward-facing layout that the user interacts with. It takes complex numeric data from the backend and translates it into a modern, minimal, dark-themed dashboard using interactive charts, timelines, and progress indicators.

Architectural Layer	Recommended Technology	Practical Implementation Role
User Interface	Next.js (TypeScript) + Tailwind CSS	Renders interactive dark-mode interfaces and tracking metrics.
Application Backend	FastAPI (Python)	High-performance asynchronous routing engine linking database entities.
Queue Management	Redis + Celery	Handles background tasks for data collection and API fetching.
Storage Solution	PostgreSQL	Handles data relationships, user tables, and analytical logs.
Linguistic Foundations	[NLTK] & [SpaCy]	Cleans text, tokenizes strings, and processes keywords.
Primary Classification	[MeaningCloud API]	Extracts sentiment trends and text categories out of the box.
Secondary Classification	[Google Cloud Natural Language AI]	Identifies specific entities and linguistic elements.

Prototyping Sandbox	[Jupyter Notebook]	Used to build and test the masculinity dictionary before deploying.
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6. Targeted User Interface Layout (The "Wrapped" Concept)

The interface avoids traditional admin structures in favor of a modern, dark-mode design focused on scannability and clear visual hierarchy.



Milestone 0: Foundational Research & Prototyping (Weeks 1–2)

- **Objective:** Formulate your underlying linguistics models and set up the local processing database.
- **Deliverable:** A verified, testable keyword dictionary built inside **[Jupyter Notebook]** and structured core application tables.

Milestone 1: Data Ingestion & Normalization Framework (Weeks 3–4)

- **Objective:** Link the live internet APIs and build the core text analysis engine.
- **Deliverable:** A running dashboard backend that successfully fetches social media text, converts it via the **Normalization Layer**, and classifications it using **[MeaningCloud]** and **[Google Cloud NLP]**. This gives you a working prototype for your advisor by the **third week of July**.

Milestone 2: Analytics Interface & Research Export Features (Weeks 5–6)

- **Objective:** Polish the consumer dashboard display and implement research logging.
- **Deliverable:** A polished visual interface showing metrics and timeline trends, complete with a clean data-export panel allowing researchers to extract anonymous statistics for publication.